

1. Cross-Validation

1. Why CV?

Goal: estimate test error without using the test set.

Training error is optimistic because the model was fit on the same data.

Workflow: tune hyperparameters on validation, then refit final model on all training data.

2. Validation Set

Method: split once, train on S_{train} , validate on S_{val} .

$$\hat{R}_{\text{val}} = \frac{1}{|S_{\text{val}}|} \sum_{(x,y) \in S_{\text{val}}} (y - \hat{f}_{\text{train}(x)})^2$$

Pros: simple, fast.

Cons: wastes data, high variance, pessimistic for the final model trained on all data.

3. k-Fold CV

Partition $S = S_1 \cup \dots \cup S_k$. For each fold i , train on $S \setminus S_i$ and test on S_i .

$$\text{CV}_k = \frac{1}{n} \sum_{i=1}^k \sum_{(x,y) \in S_i} (y - \hat{f}_{-i}(x))^2$$

Typical choices: $k = 5$ or 10 .

4. Bias-Variance Tradeoff

Small k : more bias, less variance, faster.

Large k : less bias, more variance, slower.

Rule: choose k by sample size and compute budget.

5. LOOCV

Special case $k = n$:

$$\text{CV}_n = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{f}_{-i}(x_i))^2$$

Pros: nearly unbiased, deterministic.

Cons: high variance, expensive if done naively.

6. Linear Smoothers

A linear smoother has $\hat{y} = Hy$.

Examples: OLS, ridge regression, kernel ridge, k-NN.

Hat matrix: H depends on X , not y .

7. LOOCV Shortcut

For linear smoothers, exact LOOCV uses only H_{ii} .

$$\hat{y}_i^{-i} = \frac{\hat{y}_i - H_{ii}y_i}{1 - H_{ii}}$$

Equivalent form:

$$\text{CV}_n = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - H_{ii}} \right)^2$$

Key idea: one fit, no refitting.

8. Ridge Hat Matrix

$$H_\lambda = X(X^\top X + \lambda I)^{-1} X^\top$$

Ridge leverages: diagonal entries H_{ii} .

Computation: one ridge fit + diagonal of H .

9. GCV

Replace individual leverages by their average.

$$\text{GCV} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - \frac{\text{tr}(H)}{n}} \right)^2$$

Pros: rotation invariant, stable, often close to LOOCV.

Use when: leverages are not too heterogeneous.

2. Information Criteria

1. Big Picture

CV estimates prediction error by resampling.

Information criteria estimate prediction error from one fit by adding a complexity penalty.

Focus: choose among candidate models without repeated refits.

2. KL View

$$\text{KL}(p \parallel q) = \int p(x) \log \left(\frac{p(x)}{q(x)} \right) dx$$

AIC targets expected KL divergence from truth to fitted model.

3. AIC

For a model with k parameters:

$$\text{AIC} = -2 \log \mathcal{L}(\hat{\theta}) + 2k$$

Interpretation: training log-likelihood minus optimism correction.

Choose: minimum AIC.

4. AIC for Gaussian Regression

Up to constants shared across models:

$$\text{AIC} = n \log \left(\frac{\text{RSS}}{n} \right) + 2k$$

with $\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$.

5. AIC Properties

Prediction-oriented: good for expected out-of-sample fit.

Penalty: grows with k , not directly with n .

Misspecification: still useful when the true model is not in the candidate set.

6. BIC

$$\text{BIC} = -2 \log \mathcal{L}(\hat{\theta}) + k \log n$$

Interpretation: stronger penalty than AIC for large n .

Use: favors simpler models, often consistent for selecting the true model when it is in the class.

3. Practical Selection

1. Nested CV

Avoid using the same data for both model selection and final evaluation.

Outer loop: estimate performance.

Inner loop: tune hyperparameters.

Result: less selection bias.

2. Common Pitfalls

- Do not report validation error as final test performance.
- Do not tune hyperparameters on the test set.
- Standardize features before ridge/lasso.
- LOOCV is not always best: low bias, high variance.
- Ridge shrinks coefficients but does not zero them out.

3. Quick Comparison

Validation set: cheapest, highest variance.

5/10-fold CV: good default, balanced bias/variance.

LOOCV: nearly unbiased, expensive, high variance.

AIC: fast, prediction-focused.

BIC: stronger penalty, model selection / parsimony.